AWS CERTIFIED CLOUD PRACTITIONER · CLF-C02

The essential edition

Everything the exam tests, explained just enough to understand it, without repetition. Built on the official AWS exam guide v1.3 and the AWS documentation.

4 domains
following the official task statements 1.1 to 4.3

Concise
clear explanations, decision tables and key diagrams

Exam ready
traps, key-word map and 30 self-test questions

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How to use this edition

Every topic of the official AWS exam guide v1.3 (task statements 1.1 to 4.3), explained just enough to understand it, without repetition. Each section follows one official task statement. Read one part per sitting, then do the self-test in Part 5.

> **ANALOGY**
> An everyday comparison to make an idea stick.

> **NOTE**
> A term explained in more depth.

> **ON THE EXAM**
> How questions are asked, and the traps.

The exam

Questions	65: 50 scored + 15 unscored that you cannot identify. Multiple choice (1 of 4) and multiple response (2 or more of 5 or more)
Time	90 minutes verify
Score	100 to 1000, **pass with 700**. Global score: you can be weaker in one domain. No penalty for wrong answers: never leave a blank
Delivery	Pearson VUE test center or online proctored. Valid 3 years

Domain	Weight
1 Cloud Concepts	24%
2 Security and Compliance	30%
3 Cloud Technology and Services	34%
4 Billing, Pricing, and Support	12%

The exam does not ask you to configure anything. It asks **which service or concept fits a situation**.

PART 1 · DOMAIN 1

Cloud Concepts

24% of the scored content

What the cloud is and why it is useful, good architecture, migration, and economics.

1.1 The benefits of the AWS Cloud

What cloud computing is

A **server** is a computer that runs all the time and answers requests from other computers (a website, an app). Before the cloud, companies ran their servers **on-premises**: they bought the hardware, housed it in a **data center** (power, cooling, network), hired staff to maintain it, and had to **guess** their capacity years ahead, paying everything **upfront**.

AWS's official definition: **cloud computing is the on-demand delivery of IT resources over the internet with pay-as-you-go pricing.**

- **On-demand**: resources in minutes, when you ask.
- **IT resources**: compute (processing), storage, databases, networking, and more than 200 services.
- **Over the internet**: managed remotely; the data centers belong to AWS.
- **Pay-as-you-go**: pay only for what you use, no upfront cost; stop using, stop paying.

> ANALOGY
> On-premises is a generator in your garden; the cloud is the electricity grid: you plug in and pay for what you consume.

Virtualization makes it possible: a **hypervisor** splits one physical server into several isolated **virtual machines**, so many customers share hardware safely (**multi-tenancy**). Ways to run code, from most to least management: **dedicated server** → **virtual machine** → **container** (an app packaged with its dependencies, sharing the host OS) → **function / serverless** (only your code, run on events).

IaaS, PaaS, SaaS

The three models differ by **how much the provider manages for you**.

Layer	On-premises	IaaS	PaaS	SaaS
Application	You	You	You	Provider
Data	You	You	You	Provider (you own your data)
Runtime, middleware	You	You	Provider	Provider
Operating system	You	You	Provider	Provider
Virtualization, hardware	You	Provider	Provider	Provider

- **IaaS (Infrastructure as a Service)**: virtual servers, storage, networks. You manage the OS and everything above. Most control. **Amazon EC2**, Amazon VPC, Amazon EBS.
- **PaaS (Platform as a Service)**: you bring only code and data; the provider runs the platform. **AWS Elastic Beanstalk**, Amazon RDS, AWS Lambda.
- **SaaS (Software as a Service)**: a finished application you just use. Gmail, Microsoft 365; Amazon WorkSpaces, Amazon Connect.

> ANALOGY
> Pizza: home-made (on-premises), frozen base you finish (IaaS), delivery (PaaS), restaurant (SaaS).

Deployment models

- **Cloud** (public cloud, cloud-native): everything on AWS. "Public" means shared infrastructure, not public data.
- **On-premises** (private cloud): everything in your own data center.
- **Hybrid**: on-premises and cloud **connected**, for example a bank keeping its core system on site and its mobile app on AWS. Hybrid enablers: **AWS Direct Connect**, **AWS VPN**, **AWS Storage Gateway**, **AWS Outposts** (AWS hardware in your data center).

The six advantages of cloud computing

1. **Trade fixed expense for variable expense**: no upfront investment (CapEx); pay monthly for usage (OpEx).
2. **Benefit from massive economies of scale**: AWS buys at huge volume, so prices are lower.
3. **Stop guessing capacity**: add or remove capacity in minutes as demand changes.
4. **Increase speed and agility**: resources in minutes; experiment cheaply.
5. **Stop spending money running and maintaining data centers**: focus on customers, not on hardware ("undifferentiated heavy lifting").
6. **Go global in minutes**: deploy close to users worldwide.

Key words

- **High availability**: the system stays reachable when a component fails; **no single point of failure**. On AWS: run in **several Availability Zones**.
- **Fault tolerance**: the system keeps working **without interruption** when a component fails (everything duplicated).
- **Scalability**: ability to handle more load. **Vertical** (scale up): a bigger server, with a limit. **Horizontal** (scale out): more servers, the cloud way.
- **Elasticity**: **automatic** scaling **in and out** with demand, so you never pay for idle capacity. Provided by **EC2 Auto Scaling**.
- **Agility**: speed to experiment and deliver.
- **Durability** (data is not lost; S3: 11 nines) vs **availability** (data is reachable now; S3 Standard: 99.99%).

> ON THE EXAM
> "Automatically add and remove resources" = elasticity = Auto Scaling. "No single point of failure" = high availability = multiple AZs. "Pay for use instead of buying hardware" = trade fixed expense for variable expense.

1.2 Design principles: the Well-Architected Framework

The **AWS Well-Architected Framework** is AWS's set of best practices to build good cloud systems (**workloads**), organized in **six pillars**.

Pillar	Question it answers	Key design principles
Operational excellence	Do we run and improve the system well?	Perform **operations as code**; make **frequent, small, reversible changes**; anticipate failure; **learn from operational failures**
Security	Are data and systems protected?	**Strong identity foundation** (least privilege); **enable traceability**; security **at all layers**; automate security; **protect data in transit and at rest**; prepare for security events
Reliability	Does it recover and meet demand?	**Automatically recover from failure**; **test recovery procedures**; **scale horizontally**; **stop guessing capacity**; manage change through automation
Performance efficiency	Do we use the right resources efficiently?	**Democratize advanced technologies** (use managed services); **go global in minutes**; **use serverless architectures**; experiment more often
Cost optimization	Are we wasting money?	**Adopt a consumption model**; measure efficiency; **stop spending on undifferentiated heavy lifting**; analyze and attribute expenditure

Pillar	Question it answers	Key design principles
Sustainability	Is the environmental impact minimal?	Understand impact; **maximize utilization**; use managed services and efficient hardware

General design principles: stop guessing capacity, test at production scale, automate experimentation, allow evolutionary architectures, drive decisions with data, practice with **game days**.

AWS Well-Architected Tool: free console tool to review a workload pillar by pillar. **AWS Architecture Center**: reference architectures.

> ON THE EXAM
> Pick the pillar by the main worry: losing service = Reliability; paying too much = Cost optimization; slowness = Performance efficiency; attacks or leaks = Security; daily operations = Operational excellence; energy = Sustainability.

1.3 Migration to the AWS Cloud

The AWS Cloud Adoption Framework (CAF)

The CAF prepares the **whole organization** for the cloud (people, processes, governance), not just the technology. **Six perspectives**:

Perspective	Focus	People
Business	Cloud investments serve business goals	CEO, CFO, business leaders
People	Culture, skills, training, organizational change	HR, managers
Governance	Manage program, risk, budget, benefits	CIO, program managers
Platform	Build the cloud platform and architecture	CTO, architects, engineers
Security	Confidentiality, integrity, availability of data	CISO, security teams
Operations	Run and support cloud services	IT operations

The first three are **business** capabilities, the last three **technical**. **CAF benefits** (exact guide wording): **reduced business risk**, **improved environmental, social and governance (ESG) performance**, **increased revenue**, **increased operational efficiency**.

The 7 Rs of migration

Strategy	Meaning	Example
Retire	Switch off what nobody needs	Old reporting tool

Strategy	Meaning	Example
Retain	Keep on-premises for now	Legal constraints, not ready
Rehost ("lift and shift")	Move as is, **fastest**, no code change	Servers copied to EC2
Relocate	Move a whole platform unchanged	VMware to VMware Cloud on AWS
Replatform ("lift, tinker and shift")	Small optimizations, same architecture	Database moved to **Amazon RDS**
Repurchase ("drop and shop")	Replace with another product, often SaaS	In-house CRM to SaaS CRM
Refactor / re-architect	Rewrite as cloud-native, **most effort and most benefit**	Monolith to serverless

Migration tools

- **AWS Application Discovery Service**: inventories on-premises servers and dependencies.
- **Migration Evaluator** NEW: builds a **business case** (estimated AWS cost).
- **AWS Application Migration Service**: **rehosts** servers with continuous replication and a short cutover.
- **AWS Database Migration Service (DMS)**: migrates databases with **minimal downtime**; the source keeps running and changes are replicated (**database replication**); same or different engines.
- **AWS Schema Conversion Tool (SCT)**: converts the **schema** when the engine changes (Oracle → Aurora). SCT converts the structure, DMS moves the data.
- **AWS Migration Hub**: one dashboard to track migrations.

> ON THE EXAM
> Fastest, no changes = Rehost. Database to a managed service = Replatform. Replace with SaaS = Repurchase. Cloud-native rewrite = Refactor. Business case = Migration Evaluator. Training staff = CAF People perspective.

1.4 Cloud economics

- **Fixed vs variable costs**: on-premises costs are mostly **fixed** (**CapEx**: bought upfront, used or not). Cloud costs are **variable** (**OpEx**: follow usage). Less financial risk, no big upfront investment.
- **Costs of on-premises** (total cost of ownership): hardware and its replacement, data center space, **power and cooling**, physical security, **staff**, licenses, unused spare capacity, slow procurement.

- **Licensing**: **license included** (the AWS price contains the software license, for example Windows on EC2) vs **BYOL, bring your own license** (reuse licenses you own; licenses tied to sockets or cores may need **Dedicated Hosts**; **AWS License Manager** tracks usage).
- **Rightsizing**: match instance type and size to real usage (a server 10% used is oversized). Tools: **AWS Compute Optimizer**, **Cost Explorer** recommendations.
- **Automation**: fewer errors, faster, consistent, resources only when needed (CloudFormation, Auto Scaling).
- **Economies of scale**: AWS's size lowers its costs and its prices.

> **ON THE EXAM**
> Avoid upfront investment = variable expense. Reuse per-core licenses = BYOL + Dedicated Hosts. Underused server = rightsizing.

PART 2 · DOMAIN 2

Security and Compliance

30% of the scored content

Who protects what, compliance and encryption, access control, and the security services.

2.1 The shared responsibility model

- **AWS is responsible for security OF the cloud**: physical security of data centers, hardware, global network, virtualization layer, and the software of managed services.
- **The customer is responsible for security IN the cloud**: **customer data**, **identity and access management**, guest **OS and application patches** on EC2, **network and firewall configuration** (security groups), **encryption** choices.

> **ANALOGY**
> Renting an apartment: the landlord maintains the building and its main door; you lock your own door, choose who has a key and protect what is inside.

The split **moves with the service**: the more managed, the more AWS does.

Service	AWS takes care of	The customer takes care of
Amazon EC2 (IaaS)	Data center, hardware, hypervisor	**Guest OS and patches**, applications, security groups, data, IAM
Amazon RDS (managed)	Plus the OS, **database engine installation and patching**, automated backups	Data, database users, network access, encryption settings
AWS Lambda (serverless)	Plus servers, OS, runtime, scaling	Code, permissions, data
Amazon S3	Infrastructure, durability	Access (bucket policies, IAM), encryption choice, data

Shared controls (each party in its own layer): **patch management**, **configuration management**, **awareness and training**.

> **ON THE EXAM**
> Guest OS on EC2 = customer. Database engine on RDS = AWS. Physical security = AWS. Security groups = customer. Customer data = always the customer. Patch management = shared.

2.2 Security, governance and compliance concepts

Compliance

Compliance = following external rules (laws, industry standards) and **proving** it to auditors. Programs to recognize: **PCI DSS** (payment cards), **HIPAA** (US health data), **GDPR** (EU personal data), **SOC 1/2/3** (audit reports on controls), **ISO 27001** (security management), **FedRAMP** (US government cloud).

- AWS certifies **its part**. Using a certified service does **not** make your application compliant: you must configure it correctly.
- Requirements **vary by country and industry**, and **not every service is covered by every program**.
- **AWS Artifact**: free self-service portal to **download AWS compliance reports** (SOC, PCI, ISO) and **accept agreements** (for example the HIPAA Business Associate Addendum).

Encryption

Encryption makes data unreadable without the **key**. Data is protected **in transit** (moving over a network, with **TLS / HTTPS**) and **at rest** (stored on disks, databases, S3).

Service	Purpose
AWS KMS	Create and control **encryption keys**; integrated with most services (often one checkbox). Default for **encryption at rest**
AWS CloudHSM	**Dedicated hardware security module**: single-tenant hardware, keys controlled only by you
AWS Certificate Manager (ACM)	Free SSL/TLS certificates, renewed automatically: **encryption in transit**

Logs, monitoring and governance

Question	Service	Records
Who did **what**, **when**?	**AWS CloudTrail**	Every **API call** (user, action, time, IP): the audit trail
How is it performing **now**?	**Amazon CloudWatch**	**Metrics**, logs, **alarms** (including **billing alarms**), dashboards
What is the configuration, is it **compliant**?	**AWS Config**	Configuration **history** and compliance **rules**

Governance services:

- **AWS Organizations**: many accounts, one bill; **service control policies (SCPs)** set the **maximum permissions** for accounts (guardrails).
- **AWS Control Tower**: automated setup of a secure multi-account environment (landing zone) with guardrails.
- **AWS Systems Manager**: operate servers at scale (patching, commands, inventory).
- **Access reports**: the IAM **credential report** (all users and the status of passwords, keys, MFA) and last-accessed information.

> ON THE EXAM
> Audit API calls = CloudTrail. CPU alarm or billing alarm = CloudWatch. Configuration changes and compliance = Config. Block a service for all accounts = SCP. Download a SOC 2 or PCI report = Artifact. AWS compliant does not mean you are compliant.

2.3 Access management

Authentication = who are you (password, MFA, keys). **Authorization** = what you may do (policies).

The root user

Created with the account; **unlimited access**. Protect it: **MFA**, strong unique password, **no access keys**, do not use it daily. **Root-only tasks**: change account settings (name, email, root password), **close the account**, **change or cancel the Support plan**, restore IAM permissions when all admins are locked out, register as a seller in the **Reserved Instance Marketplace**, enable **MFA Delete** on S3, sign up for GovCloud.

IAM

AWS Identity and Access Management (IAM): free, global service that controls **who can do what on which resources**.

Element	What it is
IAM user	One person or application with **long-term** credentials (password, access keys)
IAM group	A set of users sharing permissions
IAM role	An identity with **temporary credentials** that is **assumed** by AWS services (an EC2 instance reading S3), other accounts (cross-account) or federated users. Safer than keys
IAM policy	A **JSON** document: **Effect** (Allow/Deny), **Action**, **Resource**, optional Condition

- **Managed policies** (AWS managed or customer managed, reusable) vs **inline** policies (for one identity).
- **Evaluation**: everything denied by default → an **Allow** permits → an **explicit Deny always wins**.
- **Least privilege**: grant only the permissions needed. Tools: IAM Access Analyzer, last-accessed information, credential report.

Credentials and identities

- **Password policy**: length, complexity, expiration, no reuse.
- **MFA**: password + a device: **virtual MFA** app, **hardware token**, **FIDO security key / passkey**.
- **Access keys** (access key ID + secret) for CLI and SDKs: never share or put in code, rotate, never for root; **prefer roles**.
- **Secrets**: **AWS Secrets Manager** (stores and **automatically rotates** passwords and API keys) and **Systems Manager Parameter Store** (configuration values and secrets).

Who signs in	Service
Employees, many AWS accounts and apps, one login	**IAM Identity Center** (single sign-on, formerly AWS SSO)
Existing corporate identities (Active Directory, Entra ID, Okta)	**Federation** (SAML 2.0, OIDC)
An AWS service or application	**IAM role**
A user from another AWS account	**Cross-account IAM role**
Customers of your web or mobile app	**Amazon Cognito**
Managed Microsoft Active Directory	**AWS Directory Service**

Zero trust: never trust by default, always verify; identity is the main perimeter.

> **ON THE EXAM**
> EC2 needs S3 access = IAM role. One login for many accounts = IAM Identity Center. App customers = Cognito. Rotate DB passwords = Secrets Manager. Team permissions = group. Close account or change Support plan = root user.

2.4 Security components and resources

Service	Job	Cue words
AWS Shield	**DDoS** protection (floods of traffic). **Standard: free and automatic**. **Advanced: paid**, 24/7 response team, cost protection	DDoS

Service	Job	Cue words
AWS WAF	Web application firewall (layer 7): blocks **SQL injection**, **cross-site scripting**, bad IPs, bots	web exploits
AWS Firewall Manager	Applies WAF, Shield Advanced and security group rules **across all accounts**	central rules
Amazon GuardDuty	**Threat detection** with ML on CloudTrail, VPC Flow Logs and DNS logs	malicious activity, compromised
Amazon Inspector	**Vulnerability scanning** of EC2, container images, Lambda (CVEs, open ports)	vulnerabilities
Amazon Macie	Finds **sensitive data (PII) in S3**	personal data, GDPR
Amazon Detective	**Investigates** findings to find the root cause	investigate
AWS Security Hub	**One dashboard** of findings and best-practice checks (security posture)	single view
AWS Trusted Advisor	Best-practice **checks on the account**, including security (root MFA, open ports, public buckets)	recommendations

NOTE

Inspector checks your **workloads** for software vulnerabilities; **Trusted Advisor** checks your **account** for best practices (cost, performance, security, fault tolerance, limits).

- **Defense in depth**: protect every layer (data, application, compute, network, perimeter, identity, physical).
- **Third-party security products**: **AWS Marketplace**.
- **Security information**: **AWS Security Center**, **AWS Security Blog**, **AWS Knowledge Center**, documentation, re:Post.
- **Penetration testing** of your own resources on common services is allowed **without prior approval** (DoS simulation follows a separate policy).
- **Report abuse** from AWS resources: **AWS Trust & Safety team**.

ON THE EXAM

DDoS = Shield. SQL injection = WAF. Threats = GuardDuty. Vulnerabilities = Inspector. PII in S3 = Macie. Root cause = Detective. Single pane = Security Hub. Third-party firewall = Marketplace. Abuse = Trust & Safety.

PART 3 · DOMAIN 3

Cloud Technology and Services

34% of the scored content

For each service, remember **what problem it solves** and **when to choose it**.

3.1 Deploying and operating in the AWS Cloud

Every action in AWS is an **API** call. The tools are different ways to send these calls:

Method	Use
AWS Management Console	Web interface: learning, exploring, **one-time** tasks
AWS CLI	Commands in a terminal; scripts and automation
AWS SDKs	Libraries so your **application code** calls AWS (Python, Java, JavaScript...)
Infrastructure as code	Describe infrastructure in a file; AWS builds it identically every time

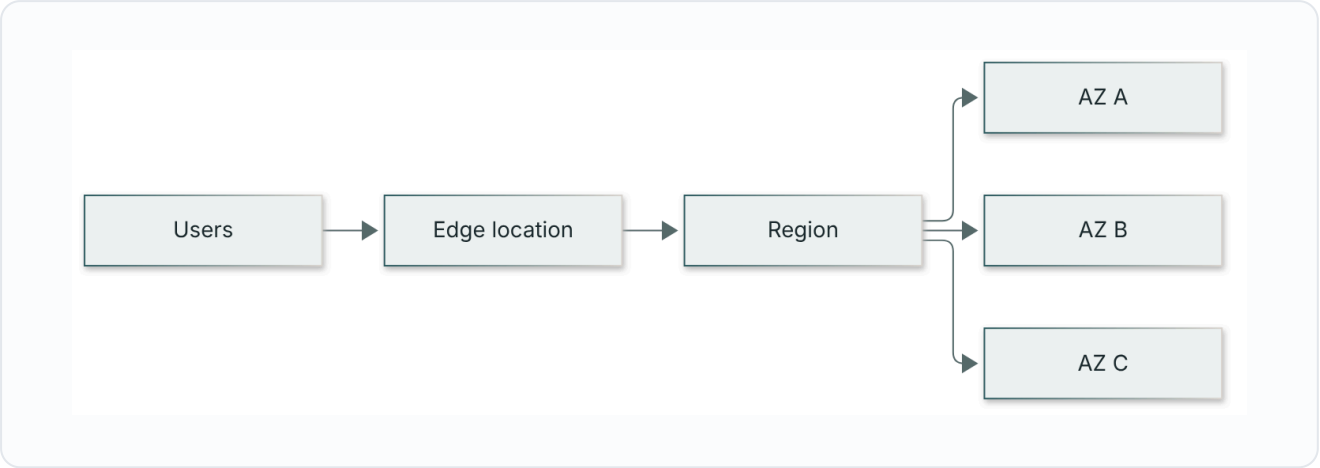
- **AWS CloudFormation**: **templates** (JSON/YAML) create resources as a **stack**; same environment in any Region or account; automatic rollback; free (pay for resources). **AWS CDK**: the same in a programming language.
- **One-time vs repeatable**: console for one-time tasks; CLI, SDKs and IaC for **repeatable**, consistent, error-free processes.
- **ARN** (Amazon Resource Name): unique identifier of a resource, used in policies.
- **Deployment models**: cloud, hybrid, on-premises (see 1.1).

> ON THE EXAM
>
> Repeatable deployments in several Regions = CloudFormation. Scripts = CLI. Code calls AWS = SDK.

3.2 The AWS global infrastructure

- **Region**: a geographic area (Frankfurt, Paris, North Virginia...), **independent and isolated**. **Data stays in the Region you choose** unless you move it.
- **Availability Zone (AZ)**: **one or more data centers** with separate power, cooling and network; each Region has several (usually 3+), kilometres apart, linked by fast private fibre. **AZs do not share single points of failure.**

- **Edge locations** (points of presence): many more sites in big cities, used by **CloudFront** (cache), **Route 53** (DNS) and **Global Accelerator**, to bring services closer to users and reduce **latency** (the delay between user and server).



- **High availability** = run in **at least two AZs** (for example servers in two AZs behind a load balancer, database with a standby in another AZ).
- **Multiple Regions** for: **disaster recovery**, **business continuity**, **low latency for users** on several continents, **data sovereignty**.
- **Choose a Region by**: compliance and data residency, proximity to users, service availability, price.
- **AWS Outposts**: AWS racks in your own data center. **AWS GovCloud (US)**: isolated Regions for US government. **AWS China Regions**: operated by local partners, separate accounts.

Disaster recovery

- **RPO** (Recovery Point Objective): how much **data** you can lose, in time.
- **RTO** (Recovery Time Objective): how long you can be **down**.

Strategy	In the second Region	Recovery	Cost
Backup and restore	Only backups	Hours	Lowest
Pilot light	Data replicated, core systems ready but off	Tens of minutes	Low
Warm standby	Small but complete copy always running	Minutes	Medium
Multi-site active/active	Full copy serving users	Near zero	Highest

AWS Backup: central backup plans across services. **AWS Elastic Disaster Recovery**: continuous replication of servers for fast recovery.

ON THE EXAM

High availability = multiple AZs. Region-wide disaster, global users, data sovereignty = multiple Regions. Cheapest DR = backup and restore; scaled-down copy running = warm standby; near-zero downtime = multi-site.

3.3 Compute services

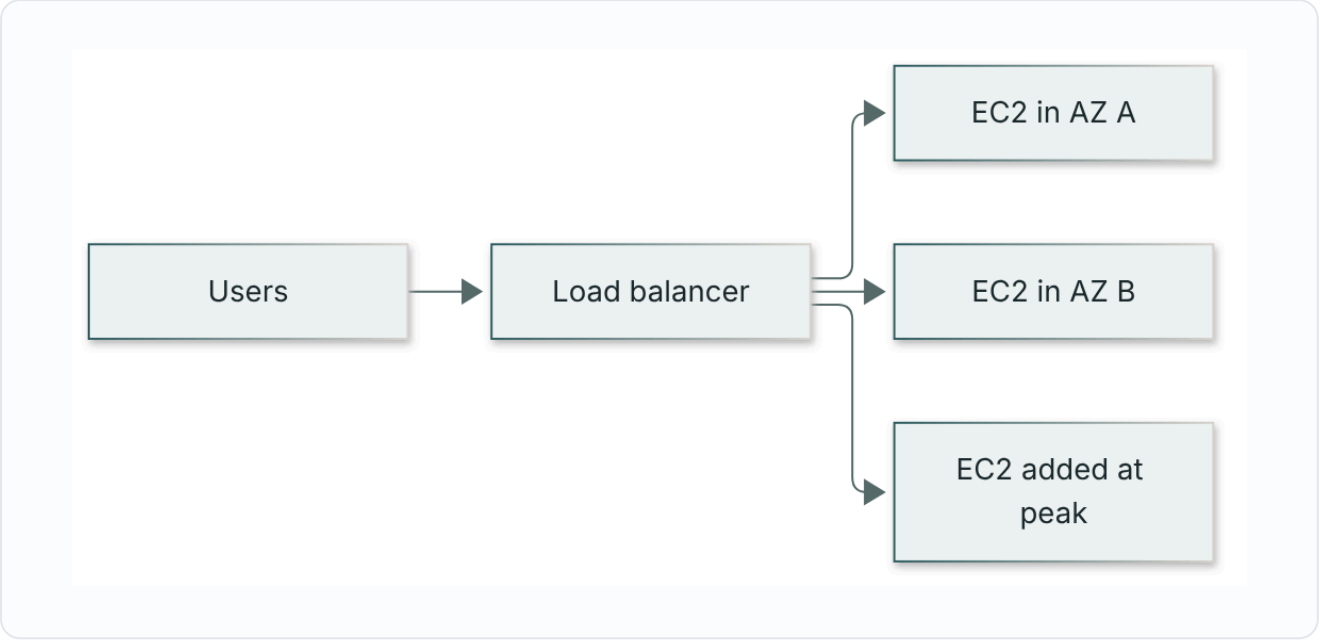
Amazon EC2

Virtual servers (**instances**) in minutes; full control of the OS (IaaS, you patch it). Launched from an **AMI** (OS template) with an **instance type** (CPU, memory), **storage** (EBS), a **security group** (firewall) and a **key pair**.

Family	Optimized for	Uses
General purpose (T, M)	Balance	Web servers, small databases
Compute optimized (C)	CPU	Batch, video encoding, gaming, scientific modeling, HPC
Memory optimized (R, X)	RAM	In-memory databases, real-time big data
Storage optimized (I, D)	Fast local storage, high **IOPS** (reads/writes per second)	Large transactional databases, data warehouses
Accelerated computing (P, G, Inf, Trn)	GPUs, ML chips	ML training, graphics

Load balancing and Auto Scaling

- **Elastic Load Balancing (ELB)**: distributes traffic across instances in **several AZs**, runs **health checks** and stops sending traffic to unhealthy instances. **Application Load Balancer** (HTTP/HTTPS, layer 7, routes by URL), **Network Load Balancer** (TCP/UDP, layer 4, extreme performance, static IP), **Gateway Load Balancer** (third-party appliances).
- **Amazon EC2 Auto Scaling**: adds instances when demand rises and **removes** them when it falls, between a **minimum** and a **maximum**; replaces unhealthy instances. **Auto Scaling provides elasticity.**



Containers and serverless

- **Container**: an application packaged with its dependencies, runs the same everywhere, lighter than a VM (Docker). Run them with **Amazon ECS** (AWS's orchestrator) or **Amazon EKS** (managed **Kubernetes**); store images in **Amazon ECR**.
- **AWS Fargate**: **serverless containers**: run containers on ECS/EKS without managing servers.
- **AWS Lambda**: **serverless functions** run on **events** (S3 upload, API request, schedule); no servers, automatic scaling, pay per request and millisecond; **up to 15 minutes** per run.

Other compute

- **AWS Elastic Beanstalk**: **PaaS**: upload code, AWS creates and manages servers, load balancer, scaling.
- **Amazon Lightsail**: simple servers at a **fixed low monthly price**, for simple sites and beginners.
- **AWS Batch**: runs large numbers of **batch jobs**.

If the question says	Choose
Full control of the OS	**EC2**
Distribute traffic, health checks	**Elastic Load Balancing**
Elasticity, add/remove instances automatically	**EC2 Auto Scaling**
Containers / Kubernetes / image registry	**ECS / EKS / ECR**
Containers without servers	**Fargate**
Code on events, no servers	**Lambda**
Upload code, AWS deploys it	**Elastic Beanstalk**

If the question says	Choose
Simple server, fixed price	**Lightsail**

3.4 Database services

- **Relational (SQL)**: tables with rows and columns linked by relations, strict schema, **transactions**; for orders, payments, accounting.
- **NoSQL**: flexible formats: **key-value**, **document** (JSON), **graph** (relationships), **in-memory**; massive scale.
- **EC2-hosted database**: full control but you patch, back up and scale. **Managed database**: AWS handles it; usually the expected answer.

Need	Service
Managed relational database (MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Db2)	**Amazon RDS**
MySQL/PostgreSQL compatible, high performance, 6 copies across 3 AZs	**Amazon Aurora**
Serverless NoSQL key-value, **millisecond** latency at any scale	**Amazon DynamoDB**
In-memory cache, microseconds, reduce database load	**Amazon ElastiCache**
Document database, **MongoDB compatible**	**Amazon DocumentDB** NEW
Graph: social networks, recommendations, fraud	**Amazon Neptune**
Data warehouse: analytics on huge historical data	**Amazon Redshift**

NOTE

OLTP = many small transactions (RDS, Aurora, DynamoDB). **OLAP** = big analytical queries over years of data (Redshift).

- **RDS Multi-AZ** = **high availability**: synchronous standby in another AZ, automatic failover.
- **Read replicas** = **performance**: copies that serve read queries.
- **Migration**: **DMS** moves data with minimal downtime; **SCT** converts the schema between engines.

ON THE EXAM

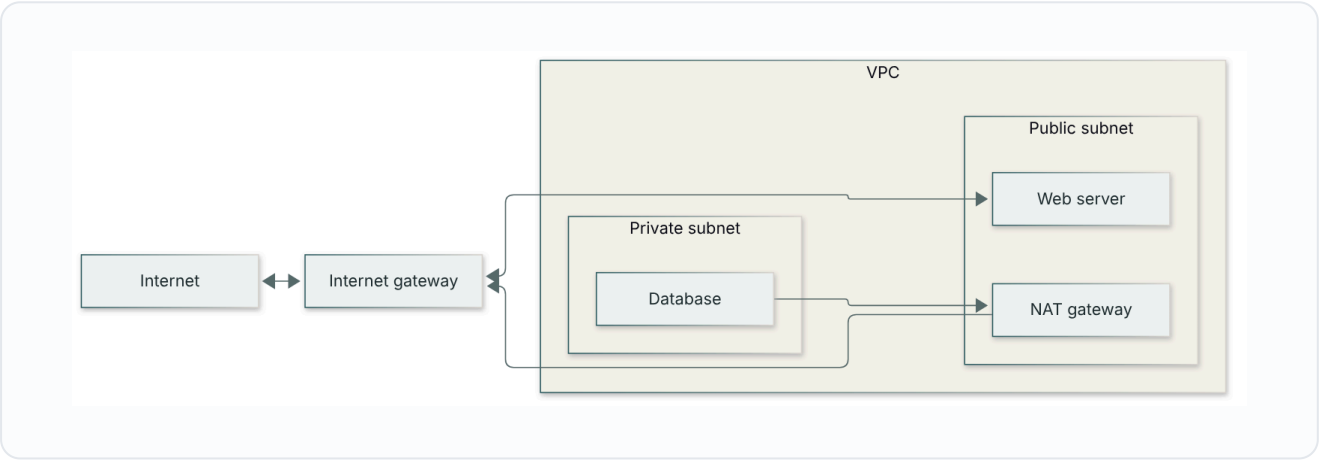
Managed SQL = RDS. Highest-performance MySQL/PostgreSQL = Aurora. NoSQL milliseconds = DynamoDB. Cache = ElastiCache. MongoDB = DocumentDB. Graph = Neptune. Warehouse = Redshift. High availability = Multi-AZ; read performance = read replicas.

3.5 Network services

Amazon VPC

Your own **logically isolated network** in a Region (free).

- **Subnets**: sections of the VPC in **one AZ**. **Public** (route to the internet gateway) for web servers and load balancers; **private** for databases.
- **Internet gateway**: connects the VPC to the internet.
- **NAT gateway**: lets **private** subnets reach the internet (updates) without being reachable from it.
- **Route tables**: decide where traffic goes.



Security groups and network ACLs

	Security group	Network ACL
Protects	An **instance**	A **subnet**
Rules	**Allow only**	**Allow and deny**
State	**Stateful**: replies allowed automatically	**Stateless**: in and out checked separately
Typical use	Allow HTTPS, SSH from my IP	**Block one malicious IP**

Amazon Inspector also reports unintended network exposure of instances.

Connecting networks

- **VPC peering**: connects two VPCs. **AWS Transit Gateway** NEW: a **hub** connecting many VPCs and on-premises networks. **AWS PrivateLink** NEW: reach services **privately**, without the internet.
- **AWS Site-to-Site VPN**: **encrypted tunnel over the internet** between your network and AWS; quick and cheap. **AWS Client VPN**: for individual users.

- **AWS Direct Connect**: **dedicated private connection**, consistent performance, lower data-out cost, weeks to set up, **not encrypted by default**.

Reaching users

- **Amazon Route 53**: **DNS** (turns names into IP addresses), domain registration, health checks, routing by **latency**, **geolocation**, **weight** or **failover**.
- **Amazon CloudFront**: **CDN** caching content in **edge locations** near users; integrates with Shield and WAF.
- **AWS Global Accelerator**: **static IP addresses** and fast routing over the AWS network, for any TCP/UDP traffic.
- **Amazon API Gateway**: create and secure APIs (often in front of Lambda).

> **ON THE EXAM**
>
> Isolated network = VPC. Private subnet needs updates = NAT gateway. Block an IP = network ACL. Many VPCs = Transit Gateway. Private access = PrivateLink. Encrypted over internet = VPN. Dedicated line = Direct Connect. DNS = Route 53. Cache near users = CloudFront. Static IPs = Global Accelerator.

3.6 Storage services

Type	How it works	AWS service
Block	A disk attached to one server	**Amazon EBS**, instance store
File	Shared folders over the network, many servers	**Amazon EFS**, **Amazon FSx**
Object	Files stored as objects with a key, reached over HTTP	**Amazon S3**

Amazon S3

Objects (0 bytes to **5 TB** each) in **buckets** (globally unique names, one Region). Virtually unlimited, **11 nines durability** (copies across 3+ AZs), **private by default**, **versioning**, static website hosting. Uses: backups, archives, media, logs, data lakes.

Class	Use	Note
S3 Standard	Frequent access	Default
S3 Intelligent-Tiering	**Unknown or changing** access	Moves objects automatically
S3 Standard-IA	Infrequent, immediate access	Retrieval fee, 30-day minimum
S3 One Zone-IA	Infrequent, **re-creatable** data	**One AZ only**
S3 Glacier Instant Retrieval	Archive, milliseconds access	90-day minimum

Class	Use	Note
S3 Glacier Flexible Retrieval	Archive, minutes to hours	90-day minimum
S3 Glacier Deep Archive	Long-term retention, almost never read	**Cheapest**, 12 to 48 h, 180-day minimum

Lifecycle policies move objects to cheaper classes or delete them automatically as they age.

Block, file, hybrid, backup

- **Amazon EBS**: persistent virtual disk for **one EC2 instance in one AZ**; **snapshots** stored in S3.
- **Instance store**: very fast disk on the host, **temporary**: lost when the instance stops.
- **Amazon EFS**: **shared Linux file system** (NFS) for many instances across AZs; grows automatically.
- **Amazon FSx**: **FSx for Windows File Server** (SMB, Active Directory), FSx for NetApp ONTAP, FSx for OpenZFS.
- **AWS Storage Gateway**: on-premises applications use cloud storage with a **local cache**: **File**, **Volume** and **Tape** gateways.
- **AWS Backup**: central backup plans across EC2, EBS, RDS, DynamoDB, EFS, FSx and more.

> ON THE EXAM
> Unlimited files, backups = S3. Unknown access = Intelligent-Tiering. Cheapest archive = Glacier Deep Archive. Re-creatable = One Zone-IA. Disk for one instance = EBS. Temporary = instance store. Shared Linux files = EFS. Windows shares = FSx for Windows. On-premises + cloud storage = Storage Gateway. Central backups = AWS Backup.

3.7 AI/ML and analytics services

AI = machines doing tasks needing human intelligence; **machine learning** = learning patterns from data to build a model; **generative AI** = creating new content.

Service	Does
Amazon SageMaker AI NEW	Build, train and deploy **your own ML models**
Amazon Q NEW	**Generative AI assistant** for business data and developers
Amazon Lex	**Chatbots** (voice and text)
Amazon Polly	**Text to speech**
Amazon Transcribe	**Speech to text**
Amazon Translate	**Translation**

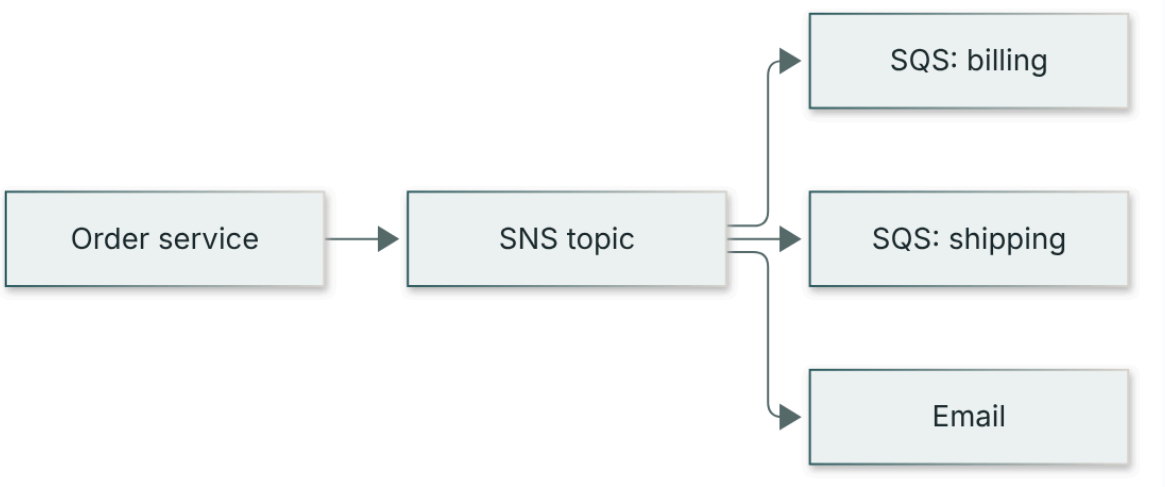
Service	Does
Amazon Comprehend	Understand text: **sentiment**, entities, key phrases
Amazon Rekognition	Analyze **images and videos**
Amazon Textract	Extract text and data from **scanned documents**

Analytics service	Does
Amazon Kinesis	**Real-time streaming** data
AWS Glue	Serverless **ETL** and data catalog
Amazon Athena	**SQL directly on files in S3**, serverless
Amazon EMR	Big data clusters (Spark, Hadoop)
Amazon Redshift	Data warehouse
Amazon OpenSearch Service	Search and log analytics
Amazon Quick NEW (formerly QuickSight)	**BI dashboards**

3.8 Other in-scope services

Application integration

- **Amazon SQS**: **message queue**; the consumer **pulls** messages; **decouples** components.
- **Amazon SNS**: **publish/subscribe**; one message **pushed** to many subscribers (email, SMS, push, SQS, Lambda); alerts and notifications.
- **Amazon EventBridge**: **event bus** routing events by rules; **scheduled** tasks.
- **AWS Step Functions**: orchestrates **workflows** of several steps.



Business applications, end-user computing, web and mobile, IoT, developer tools

Service	Does
Amazon Connect	Cloud **contact center**
Amazon SES	Sends **emails** (transactional and marketing). SNS sends notifications; SES sends real emails
Amazon WorkSpaces	**Virtual desktops**
Amazon AppStream 2.0	**Streams one application** to a browser
Amazon WorkSpaces Secure Browser NEW	Secure browser access to internal sites and SaaS
AWS Amplify	Build and host **web and mobile** apps
AWS IoT Core	Connect and manage **IoT devices**
AWS CodeBuild	Build and test code
AWS CodePipeline	Automate the release pipeline (**CI/CD**)
AWS X-Ray	**Trace** requests to troubleshoot distributed apps

Management and governance

Service	Does
AWS Service Catalog	Catalog of **approved** products employees can launch
AWS License Manager	Track licenses (BYOL)
AWS Compute Optimizer	Rightsizing recommendations
AWS Health Dashboard	Service status and events affecting your resources
Service Quotas NEW	View and raise service limits
AWS Systems Manager	Operate servers at scale
AWS Organizations / Control Tower	Multi-account management / governed multi-account setup

Look-alike services

They look alike	The difference
CloudTrail / CloudWatch / Config	Who did what / performance and alarms / configuration and compliance
SQS / SNS / EventBridge	Queue (pull) / push to many / event routing and schedules
GuardDuty / Inspector / Trusted Advisor	Threats / vulnerabilities / account best practices

They look alike	The difference
Shield / WAF / Firewall Manager	DDoS / web exploits / central rules
KMS / CloudHSM / ACM / Secrets Manager	Keys / dedicated key hardware / TLS certificates / rotated secrets
S3 / EBS / EFS / instance store	Objects / one-instance disk / shared Linux files / temporary disk
RDS / DynamoDB / Redshift	Relational / NoSQL / warehouse
Multi-AZ / read replica	Availability / read performance
CloudFront / Global Accelerator	Caching content / static IPs and routing
Direct Connect / VPN	Private line / encrypted tunnel over internet
IAM Identity Center / Cognito	Employees / app customers
Beanstalk / CloudFormation / Lightsail	Deploy code / infrastructure as code / simple server
WorkSpaces / AppStream 2.0	Full desktop / one application
Macie / GuardDuty	PII in S3 / threats

PART 4 · DOMAIN 4

Billing, Pricing, and Support

12% of the scored content

How AWS charges, cost tools, and Support.

4.1 Pricing models

Principles: **pay as you go**; **pay less when you reserve** (1 or 3 years); **pay less per unit when you use more** (volume tiers). You pay for **compute**, **storage** and **data transfer out**. About **730 hours** in a month: hourly price × 730 = monthly cost.

Option	Commitment	Discount	Best for
On-Demand	None	None	Short, unpredictable, dev/test, cannot be interrupted
Savings Plans	1 or 3 years of **$/hour spend**	Up to ~72%	Steady usage with **flexibility**. **Compute Savings Plans** cover any instance family, size, Region, OS, plus **Fargate and Lambda**
Reserved Instances	1 or 3 years, specific configuration	Up to ~72%	Steady workloads. **Standard** (largest discount, sellable in the **RI Marketplace**) or **Convertible** (exchange family/OS). All, partial or no upfront
Spot Instances	None	Up to ~90%	**Interruptible** jobs (batch, analysis, rendering); **2-minute warning**
Dedicated Hosts	On-Demand or reserved		**BYOL** per socket/core, compliance; whole physical server
Dedicated Instances	On-Demand or reserved		Hardware dedicated to your account
Capacity Reservations	None	None alone	**Guarantee capacity in one AZ**

- **RI flexibility**: a Regional RI applies to any size in the same family; Convertible RIs can be exchanged.
- **RIs and Savings Plans in Organizations**: discounts are **shared across all accounts** (can be turned off).
- **Data transfer**: **into AWS free**; **out to the internet charged**; **between Regions charged**; between AZs small charge; same AZ private IP free. CloudFront and Direct Connect reduce egress cost.

- **Storage pricing**: amount, class, requests, retrieval fees, minimum durations.
- **Free**: IAM, VPC, Organizations, Auto Scaling, CloudFormation, Beanstalk (you pay for resources). **Free Tier**: always-free and limited-time offers (check current terms).

ON THE EXAM

Unpredictable = On-Demand. Steady and flexible (incl. Lambda/Fargate) = Compute Savings Plans. Steady specific instance = Standard RI. Change family later = Convertible RI. Cheapest interruptible = Spot. Per-core licenses = Dedicated Hosts. Guaranteed capacity in an AZ = Capacity Reservation. Data in = free.

4.2 Billing, budgets and cost management

Tool	Purpose
AWS Pricing Calculator	**Estimate** costs **before** building
AWS Budgets	**Alerts** when actual or **forecasted** cost/usage passes a threshold; can trigger actions
AWS Cost Explorer	**Visualize** past costs, **forecast**, rightsizing and RI/Savings Plans recommendations
AWS Cost and Usage Report (CUR)	**Most detailed** line-item data, delivered to **S3**
Cost allocation tags	Key/value labels to split costs by project or team; **AWS-generated** and **user-defined**; **must be activated**
AWS Organizations, consolidated billing	**One bill**, usage **combined** for volume discounts, shared RI/SP discounts; free
AWS Cost Anomaly Detection	Alerts on unusual spending
AWS Price List API	Prices programmatically

Billing and account questions go to the **Support Center** (included in every plan). Create a budget on day one.

ON THE EXAM

Estimate = Pricing Calculator. Alert = Budgets. Analyze and forecast = Cost Explorer. Most detailed = CUR. Cost per department = tags. One bill = consolidated billing.

4.3 Technical resources and AWS Support

	Basic	Business Support+ NEW	Enterprise Support	Unified Operations NEW
For	Everyone, free	**Minimum for production**	Business-critical	Mission-critical
Technical cases	**No**	24/7	24/7	24/7
Critical system down		< 30 min	**< 15 min**	**< 5 min**
Production down		< 1 h	< 1 h	< 1 h
Technical Account Manager	No	No	**Designated**	**Designated**
Trusted Advisor	Core checks	All checks	All + Priority	All + Priority
Price	Free	From ~$29/month verify	From ~$5,000/month verify	verify

Old names you may meet (ending 1 January 2027): **Developer** and **Business** → Business Support+; **Enterprise On-Ramp** → Enterprise Support; **Enterprise** = Enterprise Support.

- **Technical Account Manager (TAM)**: designated technical expert giving proactive guidance (Enterprise tiers).
- **AWS Trusted Advisor**: checks in **cost optimization**, **performance**, **security**, **fault tolerance**, **service limits** (and operational excellence).
- **AWS Health Dashboard**: service status and **events affecting your resources**; **AWS Health API** to automate.
- **Resources**: documentation, whitepapers, blogs; **AWS Prescriptive Guidance** (migration patterns); **AWS Knowledge Center** (frequent support questions); **AWS re:Post** (community Q&A); **AWS Support Center** (cases); **AWS Trust & Safety** (report abuse).
- **AWS Partner Network**: **ISVs** sell software; **system integrators / consulting partners** deliver services. Partner benefits: training and certification, events, volume discounts.
- **AWS Marketplace**: catalog of third-party software billed on your AWS bill; cost management, governance and entitlement.
- **AWS Professional Services** (AWS experts for projects) and **AWS solutions architects** (architecture guidance).
- **SLAs**: availability commitments with service credits.

ON THE EXAM

Minimum for production = Business Support+. Designated TAM = Enterprise Support or Unified Operations. Fastest = Unified Operations. No technical cases = Basic. Community = re:Post. Frequent questions = Knowledge Center. Abuse = Trust & Safety. Third-party software = Marketplace.

PART 5

Exam day

Strategy, key words, checklist and self-test

How to read questions, the checklist for the last hour, and 30 questions to test yourself.

5.1 How to answer

1. Read the **last sentence first**: what exactly is asked (cheapest? most available? least effort?).
2. Read the scenario and underline the key words.
3. **Eliminate** the two answers that are clearly wrong, then choose between the remaining two.
4. For multiple response, select **exactly** the number requested.
5. When unsure, prefer the **managed or serverless** service: AWS likes answers where AWS does the work.
6. Never leave a blank: wrong answers cost nothing.
7. A strange, very hard question may be one of the 15 unscored ones: do your best and move on.
8. About 80 seconds per question: flag long ones and come back at the end.

5.2 Last-hour checklist

- The 6 advantages of cloud, the 6 Well-Architected pillars, the 6 CAF perspectives and 4 benefits, the 7 Rs
- Shared responsibility for EC2, RDS, Lambda, S3; root-only tasks; least privilege
- CloudTrail vs CloudWatch vs Config; the security services; security group vs network ACL
- Region, AZ, edge location; 4 factors to choose a Region; DR strategies with RPO and RTO
- S3 classes in order; EBS vs EFS vs instance store; database per need
- EC2 buying options; data transfer rules; cost tools; the 4 Support plans and the old names

5.3 Test yourself

> **NOTE**
> Cover the answers in italics, answer each question aloud, then check. Every wrong answer points to a section to re-read.

1. A startup does not want to buy hardware and wants to pay monthly for what it uses. Which advantage? *Trade fixed expense for variable expense.*
2. Which pillar includes "automatically recover from failure"? *Reliability.*
3. A company wants to move servers to AWS as fast as possible without changing them. Strategy? *Rehost.*
4. Which CAF perspective covers training and culture? *People.*
5. On Amazon EC2, who patches the operating system? *The customer.* On Amazon RDS, who patches the database engine? *AWS.*
6. An EC2 instance must read files in S3. What is the secure way? *Attach an IAM role to the instance.*
7. Which service shows who deleted an S3 bucket yesterday? *CloudTrail.*
8. An auditor wants AWS's PCI DSS report. Where? *AWS Artifact.*
9. You must block one malicious IP address for a whole subnet. What? *A network ACL with a deny rule.*
10. The website suffers SQL injection attempts. Service? *AWS WAF.*
11. Which service finds personal data stored in S3? *Amazon Macie.*
12. Employees should sign in once to reach 15 AWS accounts. Service? *IAM Identity Center.*
13. Your application must stay up if one data center fails. Design? *Instances in several Availability Zones behind a load balancer.*
14. Which DR strategy keeps a small but complete copy running in another Region? *Warm standby.*
15. Traffic varies a lot during the day. Which service adds and removes instances automatically? *EC2 Auto Scaling.*
16. Run code when a file is uploaded, without managing servers? *AWS Lambda.*
17. Logs must be kept 7 years for compliance and are almost never read. Cheapest storage? *S3 Glacier Deep Archive.*
18. Several Linux instances need the same shared files. Service? *Amazon EFS.*
19. A game needs a NoSQL database with millisecond latency at any scale. Service? *Amazon DynamoDB.*
20. Managers want sales trends over five years from huge data. Service? *Amazon Redshift.*
21. Users in Asia complain that images load slowly from your European website. Service? *Amazon CloudFront.*

22. A batch job can be interrupted and must be as cheap as possible. Purchasing option? *Spot Instances.*
23. A database runs 24/7 for the next 3 years. Cheapest option? *Reserved Instances (or a Savings Plan).*
24. Finance wants an email when the monthly forecast exceeds the budget. Tool? *AWS Budgets.*
25. Which is the minimum Support plan recommended for production workloads? *Business Support+.*
26. Which Support plans include a designated Technical Account Manager? *Enterprise Support and Unified Operations.*
27. Where do developers ask questions to the community? *AWS re:Post.*
28. Is data transferred into AWS from the internet charged? *No.*
29. Which service gives best-practice recommendations on cost, security, performance, fault tolerance and limits? *AWS Trusted Advisor.*
30. A company wants a cost estimate of its new architecture before building it. Tool? *AWS Pricing Calculator.*

Then practise with the 70 questions of this guide, the official AWS Practice Question Set (free on Skill Builder) and a free practice exam. Book the exam when you score 85% or more.